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A condition-based maintenance of a dependent degradation-threshold-shock model in a system with multiple degradation processes

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Abstract

This paper proposes a condition-based maintenance strategy for a system subject to two dependent causes of failure: degradation and sudden shocks. The internal degradation is reflected by the presence of multiple degradation processes in the system. Degradation processes start at random times following a Non-Homogeneous Poisson Process and their growths are modelled by using a gamma process. When the deterioration level of a degradation process exceeds a predetermined value, we assume that a degradation failure occurs. Furthermore, the system is subject to sudden shocks that arrive at the system following a Doubly Stochastic Poisson Process. A sudden shock provokes the total breakdown of the system. Thus, the state of the system is evaluated at inspection times and different maintenance tasks can be carried out. If the system is still working at an inspection time, a preventive maintenance task is performed if the deterioration level of a degradation process exceeds a certain threshold. A corrective maintenance task is performed if the system is down at an inspection time. A preventive (corrective) maintenance task implies the replacement of the system by a new one. Under this maintenance strategy, the expected cost rate function is obtained. A numerical example illustrates the analytical results.

Keywords: Condition based maintenance, degradation threshold shock model, doubly stochastic Poisson process, gamma process, non homogeneous Poisson process, pitting corrosion.

1. Introduction

Maintenance refers to the set of necessary operations applied to a system so that it can work properly. Nowadays, maintenance plays an important role in most companies, which try to provide high quality products but minimizing the production cost [1]. Maintenance activities performed on an industrial system can increase not only its safety, but also ensure its availability and correct functioning. Traditionally, these maintenance tasks were allocated based on requirements in legislation, company standards or in-house maintenance experience. However, in the early 1960s, authors as Barlow and Hunter, Radner and Jorgenson or McCall [2–4] started developing mathematical models, which aim to quantify costs and to find the optimum balance between the maintenance cost on one side, and the associated cost (benefit) on the other.

Maintenance strategies regulate the different maintenance tasks which will be performed on systems. In general, maintenance tasks are classified in corrective maintenance and preventive maintenance. Corrective maintenance is defined as the maintenance which is required when a system has failed. Preventative maintenance is a maintenance planned in order to prevent the occurrence of future failures. It is performed when a system is still working. Preventive maintenance activities in single-unit systems were classified by Rausand and Høyland [5] in: i) Age-based maintenance, where maintenance tasks are performed when the system exceeds a certain age. ii) Calendar-based maintenance, where maintenance tasks are performed at fixed time instants. iii) Condition-based maintenance, where maintenance tasks are based on one or several variables which measure the state of the system. The state variables are monitored continuously or on a regular basis. The maintenance of the system is performed when some state-variable exceeds a certain fixed level. Because of its properties, this kind of maintenance is widely used in the current literature (see e.g. [6–9]).

Many systems are degraded physically over time and a measurable physical deterioration almost always precedes the failure. The deterioration level of the system is represented by a degradation process. But, in practice, many systems are subject to multiple degradation processes due to which they can be degraded in more than one way. A characteristic example of this type of multiple degradation is the pitting corrosion. Pitting corrosion consists of the appearance of small pits on the surface of some metals

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and alloys. Each pit is considered as a degradation process. Pits are deteriorated in a way that when they start growing, the environment developed is such that stimulates their own growth [10]. Therefore, a pitting corrosion model is a combination of two processes: the initiation process and the growth process.

It is well established the pitting corrosion process has a stochastic nature [11, 12]. In the current literature about degradation models, several authors studied pitting corrosion, proposing different stochastic models, both for the initiation of degradation processes and for its growth. Among others, Valor *et al.* [13] proposed a model where the initiation process was distributed under a Non-Homogeneous Poisson Process (NHPP), and the growth process was simulated using a non-homogeneous Markov process.

On the other hand, Sheikh *et al.* [14] treated pitting corrosion as a time-dependent stochastic damage process characterized by an exponential or logarithmic growth. Van Noortwijk and Klatter [15] developed a maintenance model for the sea-bed protection of storm-surge barriers where the stochastic initiation process and the growth process were modelled as a Poisson process and a gamma process, respectively. Mercier *et al.* [16] also used this combination of processes in order to model the degradation process of passive components within electric power plants. In addition to this, Kuniewski *et al.* [17] and Castro *et al.* [18] provided a new approach to pitting models by combining the NHPP for the initiation process and the gamma process for the growth.

A gamma process is a stochastic process, proposed initially by Abdel-Hameed [19] as a specific model for degradation occurring randomly in time, and characterised by independent and non-negative increments distributed by using gamma distributions with identical scale parameters. Because of its properties, the gamma process is considered as one of the most appropriate processes for the stochastic modelling of monotonic degradation accumulated over time, in a succession of tiny increments such as wear, stress, corrosion, erosion or the degradation process growth. A broad survey about gamma processes was performed by van Noortwijk [20].

However, systems are not only subject to internal degradation, but also are exposed to sudden shocks that can cause their failure. Models which consider systems subject to these two competing causes of failure (degradation and shocks) are called *Degradation-Threshold-Shock* (DTS) models. As far as we know, Lemoine and Wenocur [21] were the first to combine both competing causes of failure. So, the failure time of a system subject to degradation and shocks is the minimum of the moment when degradation first reaches a critical threshold and the moment when a shock occurs. Lin *et al.* [22] proposed a DTS model where the growth process followed a continuous-time semi-Markov process and where the process of sudden shocks was distributed by using a Homogeneous Poisson process (HPP). Wang *et al.* [23] analysed a DTS model where the growth of the degradation process followed a Gaussian process and the occurrence of shocks was considered both as a fixed time period and as varying time periods under an HPP. Singpurwalla [24] and Cox [25] developed different approaches to analyse stochastic process-based models, including also DTS models.

Some authors analysed maintenance strategies applied to DTS models subject to only one degradation process, Li and Pham [26] provided a condition-based maintenance of a system where degradation was modelled as a function of time and the process of shocks was distributed by using a compound Poisson process. Huynh *et al.* [27, 28] modelled different maintenance strategies for systems where the growth of the degradation process was distributed by a gamma process and the process of sudden shocks followed an NHPP with intensity dependent on the deterioration level of the system. Asadzadeh and Azadeh [29] proposed a condition-based maintenance model describing fatigue crack degradation with a Paris-Erdogan model and random positive normal shocks with Weibull frequency model. XiaoFei and Min [30] analysed a system subject to shocks arriving following a time HPP and degradation process distributed under a typical Poisson stochastic distribution. Deloux *et al.* [31] combined statistical process control and condition-based maintenance for a system with two failure mechanisms, deterioration and shocks due to a stressful environment. They propose to model the failure process by two explicative variables a cumulative deterioration process, which following a non-decreasing stochastic process and a stress covariate, distributed under a stochastic process fluctuating around a given mean.

On the other hand, DTS models where systems subject to only one degradation process are extended by different authors, which analyse systems subject to multiple degradation processes. Castro [32] and Castro *et al.* [33] studied different maintenance strategies for systems subject to multiple degradation processes and sudden shocks. They assumed that both causes of failure (degradation and sudden shocks) are independent. Li and Pham [34] proposed a stochastic failure model for a system subject to two degradation processes and to a shock process. They suppose that the growth process was distributed as different continuous probabilistic functions and shocks under a compound Poisson process. Wang and Pham [35] provided a model for a system subject to m degradation processes described by a function of time influenced by the shock process, distributed by using an HPP.

In this paper, we propose a condition-based maintenance strategy for a system subject to multiple degradation processes and sudden shocks, assuming that both causes of failure are dependent. Degradation processes start at random times following an NHPP and grow according to a gamma process. The system fails when the deterioration level of a degradation process exceeds a certain threshold. In addition, sudden shocks arrive at the system according to a Doubly Stochastic Poisson Process (DSPP), also called Cox process. This kind of processes describes a Poisson process with a random intensity that might represent a random environment or a field that influences the Poisson locations of the points. They were first introduced by Cox [36] and later were described by Cox and Lewis [37]. Afterwards, different authors considered the shock process as a DSPP (see e.g. [38, 39]). In our model, the deterioration levels of the existing degradation processes in the system influence the sudden shock process. A sudden shock provokes the total failure of the system.

Since the continuous monitoring of the system is too costly, it is inspected each T time units. These instants are called inspection times. During inspections, the state of the system and the deterioration level of each degradation process is checked. If the system has not failed, the deterioration level of each existing degradation process in the system is measured. Let M be a certain threshold from which we consider the system is still functioning but is too deteriorated. A preventive maintenance task is performed when the deterioration level of a degradation process has exceeded this predetermined deterioration level M in an inspection time. A corrective maintenance task is performed when the system is down in the inspection time.

Each maintenance task has a fixed cost associated. An objective of this paper is to find optimal values both for the threshold M and for the time between inspections T. This implies to obtain values for M and T which minimize an objective cost function. Let $\{Z(t), t \geq 0\}$ be a regenerative process. We denote by $D_1, D_2, D_3, \ldots$ their regeneration epochs, by $L_i = D_i - D_{i-1}$, $i = 1, 2, \ldots$ the length of the i-th renewal cycle being $D_0 = 0$, by $C(t)$ the cost of the system at time t, and by C_i the total cost in the i-th renewal cycle. Therefore, if a cost structure is imposed on the regenerative process $\{Z(t), t \geq 0\}$, it is well known that (see [40] pg. 41)

$$\frac{C(t)}{t} \rightarrow \frac{E[C_1]}{E[L_1]}, \tag{1}$$

with probability one, that means, the long-run average cost per time unit is equal to the expected cost in a renewal cycle divided by the expected length of this cycle for almost any realization of the process. The long-run average cost criterion is widely used in the reliability literature [27, 28] and we shall use (1) as objective function to analyse the optimal strategy for this maintenance model.

This paper is structured as follows. In Section 2 the general framework of the model is described. Section 3 details the maintenance strategy used in this model. An illustrative example is shown in Section 4. Section 5 concludes and shows further possible extensions of this paper.

2. General framework

A maintenance model for a system subject to two competing causes of failure, internal degradation and sudden shocks, is considered. Both competing causes of failure are dependent.

2.1. Assumptions of the model

The general assumptions of this model are:

1. The system starts working at time $t = 0$. The system is subject to multiple internal degradation processes. The successive degradation processes start at random times following an NHPP $\{N_d(t), t \geq 0\}$, where $N_d(t)$ denotes the number of degradation processes in the system at time t. Therefore, $N_d(t)$ degradation processes are evolving simultaneously in the system at time t. We assume that $\{N_d(t), t \geq 0\}$ has intensity $m(t)$ and cumulative intensity $M(t)$ given by
$$M(t) = \int_0^t m(u)\, du, \quad t \geq 0,$$
where t is the age of the system and $m(t)$ is a non-decreasing function in t. The sequential initiation points of the degradation processes are denoted by $0 \leq T_1 \leq T_2 \leq \ldots$, where T_i denotes the initiation point of the i-th degradation process.
2. We assume that degradation processes grow independently each other. The deterioration level of the degradation processes is developed depending on the environment and the component conditions according to a gamma process. Let $X_i(t)$, $i = 1, 2, \ldots$ be the deterioration level of the i-th degradation process t time units after its initiation. We assume, $\{X_i(t),\ t \geq 0\}$ is distributed by a gamma process with shape and scale parameters given by α and β, respectively ($\alpha, \beta > 0$). That means, for $s < t$, the density of the deterioration increment $X_i(t) - X_i(s)$ of the i-th degradation process is given by
$$f_{\alpha(t-s),\beta}(x) = \frac{\beta^{\alpha(t-s)}}{\Gamma(\alpha(t-s))} x^{\alpha(t-s)-1} e^{-\beta x}, \quad x \geq 0, \tag{2}$$
for $i = 1, 2, \ldots$, where Γ denotes the gamma function defined as
$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u}\, du.$$
A degradation failure occurs when the deterioration level of a degradation process first exceeds a known fixed threshold L called failure threshold.
3. The system is also subject to sudden shocks occurring randomly in time, provoking the failure of the system. Sudden shocks arrive according to a counting process $\{N_s(t), t \geq 0\}$. This process shows the dependence between the competing causes of failure in the following way: each cause of failure has its own failure probability density function and there is an interaction between both of them. This means that the intensity of the process of sudden shocks at time t depends on the deterioration level of the existing degradation processes in the system. This dependence is reflected in the fact that the system

is more susceptible to a sudden shock when the deterioration level of a degradation process exceeds a certain level M_s. Let $\{J(t), t \geq 0\}$ be the covariate process $\{X_i^*(t), N_d(t), i = 1, 2, \ldots, N_d(t), t \geq 0\}$, where $N_d(t)$ is the number of degradation processes evolving in the system at time t and $X_i^*(t)$ denotes the deterioration level of the i-th degradation process at time t, that is, $X_i^*(t) = X_i(t - T_i)$ for $t \geq T_i$. Then, $\{N_s(t), t \geq 0\}$, given $J(t)$, is a Poisson process with random intensity

$$h\left(t|_{\{J(u), 0 \leq u \leq t\}}\right) = r_1(t) \prod_{i=1}^{N_d(t)} \mathbf{1}_{\{X_i^*(t) \leq M_s\}} + r_2(t) \left[1 - \prod_{i=1}^{N_d(t)} \mathbf{1}_{\{X_i^*(t) \leq M_s\}}\right], \quad (3)$$

for $t > 0$, where r_1 and r_2 denote two failure rate functions which verify $r_1(t) \leq r_2(t)$ for all $t \geq 0$. Symbol $\mathbf{1}_{\{\}}$ denotes the indicator function which equals 1 if the argument is true and 0 otherwise, by convection $\prod_{i=1}^{0} = 1$. Note that $h(t)$ can be specified for a realization of the covariate process $\{J(t), t \geq 0\}$. Hence, $\{N_s(t), t \geq 0\}$ is a DSPP with information process $\{J(t), t \geq 0\}$.

4. The system is inspected each T time units to evaluate its state. If the system is down, a corrective maintenance is performed and the system is replaced by a new one. If the system is not down at the inspection time, the deterioration level of each degradation process is checked. If the deterioration level of a degradation process exceeds a certain threshold M ($M < L$), a preventive maintenance is performed and the system is replaced by a new one. Otherwise, no maintenance task is performed. We assume the time necessary to perform a maintenance action is negligible.
5. The costs of a corrective and preventive maintenance are C_c and C_p, respectively. The cost of each inspection is C_i and the cost incurred by the inactivity of the system is C_d. We assume $C_c > C_p > C_d > C_i > 0$.

2.2. First hitting time distributions

In this model with multiple degradation processes, we assume that the system fails when one of these degradation processes first reaches the failure threshold L. Let σ_L be the first time at which the deterioration level of the i-th degradation process exceeds L since its initiation for $i = 1, 2, \ldots$ The distribution function of σ_L is given by

$$F_{\sigma_L}(t) = P\left[X_i(t) \geq L\right] = \int_L^\infty f_{\alpha t, \beta}(x)\, dx = \frac{\Gamma(\alpha t, L\beta)}{\Gamma(\alpha t)}, \quad (4)$$

for $t \geq 0$, where $f_{\alpha t, \beta}(x)$ is given by (2) and

$$\Gamma(\alpha, x) = \int_x^\infty z^{\alpha - 1} e^{-z}\, dz,$$

denotes the incomplete gamma function for $x \geq 0$ and $\alpha > 0$. The distribution function F_{σ_L} given by (4) is known as the first hitting time distribution to the failure threshold L whose density function is expressed as

$$f_{\sigma_L}(t) = \frac{\alpha}{\Gamma(\alpha t)} \int_{L\beta}^\infty \{\log(z) - \psi(\alpha t)\}\, z^{\alpha t - 1} e^{-z}\, dz, \quad (5)$$

for $t \geq 0$, where function $\psi(a)$ is called the digamma function, which corresponds to the derivative of the logarithm of the gamma function, that is

$$\psi(a) = \frac{\Gamma'(a)}{\Gamma(a)} = \frac{\partial \log \Gamma(a)}{\partial a}.$$

The threshold M is defined as the deterioration level from which the system is considered as too much worn and a preventive replacement is triggered. Let σ_M be the first time at which the deterioration level of the i-th degradation process exceeds the threshold M since its initiation for $i = 1, 2, \ldots$ The first hitting time distribution to the level M is obtained from (4) as

$$F_{\sigma_M}(t) = \frac{\Gamma(\alpha t, M\beta)}{\Gamma(\alpha t)}, \quad t \geq 0, \quad (6)$$

with density f_{σ_M} given by (5) replacing L by M.

2.3. Distribution of $\sigma_L - \sigma_M$

The probability distribution of the random variable $\sigma_L - \sigma_M$ is used in the subsequent mathematical development of this model. This distribution is difficult to derive due to the "overshoot behaviour" of the gamma process. Overshoot behaviour means that if $X_i(t)$ is the deterioration level of the i-th degradation process t time units after its initiation, then it does not exactly reach the level M but attains it with a non-degenerative random overshoot (see, e.g., [41] and [42]). This happens due to the fact that the gamma process is a jump process. Thus, $X_i(\sigma_M) \geq M$ and hence $\sigma_L - \sigma_M$ does not have the same probability distribution as σ_{L-M}. By Bertoin [43], the joint probability density of $(\sigma_M, X_i(\sigma_M))$ is

$$f_{\sigma_M, X_i(\sigma_M)}(x, y) = \int_0^\infty \mathbf{1}_{\{M \leq y < M + s\}} f_{\alpha x, \beta}(y - s) \mu(ds), \quad (7)$$

for $i = 1, 2, \ldots$, where $\mu(ds)$ denotes the Lévy measure of the gamma process with parameters α and β given by

$$\mu(ds) = \alpha \frac{e^{-\beta s}}{s}, \quad s > 0.$$

The survival function of $\sigma_L - \sigma_M$ is obtained as

$$\begin{aligned} \bar{F}_{\sigma_L - \sigma_M}(t) &= P\left[\sigma_L - \sigma_M \geq t\right] \\ &= \int_{x=0}^\infty \int_{y=M}^\infty f_{\sigma_M, X_i(\sigma_M)}(x, y) F_{\alpha t, \beta}(L - y)\, dy\, dx, \end{aligned} \quad (8)$$

where $f_{\sigma_M, X_i(\sigma_M)}(x, y)$ is given by (7) and $F_{\alpha t, \beta}$ denotes the distribution function of the density function $f_{\alpha t, \beta}$ given by (2).

2.4. Distribution of counting processes

Since the different degradation processes start at random times, a stochastic model is developed in order to find the distribution of the time required to reach the deterioration level M, resulting from the combination of the initiation process and the growth process of degradation processes. Let S_i and W_i be the time where the deterioration level of the i-th degradation process exceeds the thresholds M and L, respectively. Hence, S_i and W_i are given by

$$S_i = T_i + \sigma_M \quad \text{and} \quad W_i = T_i + \sigma_L,$$

for $i = 1, 2, \ldots$ A realization of the degradation processes is shown in Fig. 1.

Let $N_M(t)$ be the number of degradation processes whose deterioration levels exceed the preventive threshold M at time t, that is

$$N_M(t) = \sum_{i=1}^{\infty} \mathbf{1}_{\{S_i \leq t\}}. \tag{9}$$

The distribution of the counting process $\{N_M(t),\, t \geq 0\}$ is obtained from the following result.

Lemma 1. *Let $\{T_n,\, n \geq 1\}$ be a non-homogeneous Poisson process with intensity function $\lambda(t)$ and let $\{U_n,\, n \geq 1\}$ be a sequence of independent and identical distributed random variables with density function given by f. It is denoted*

$$T'_n = T_n + U_n, \quad n = 1, 2, \ldots$$

Let $\{N'(t),\, t \geq 0\}$ be the counting process associated with the random variables T'_n, that is,

$$N'(t) = \sum_{n=1}^{\infty} \mathbf{1}_{\{T'_n \leq t\}}, \quad t \geq 0.$$

Then $\{N'(t),\, t \geq 0\}$ is an NHPP with intensity $\lambda * f$.

Proof. It is provided by Kuniewski *et al.* [17] in Subsection 4.1. □

Considering Lemma 1, the counting process $\{N_M(t), t \geq 0\}$ is an NHPP with intensity

$$m_M(t) = \int_0^t m(u) f_{\sigma_M}(t-u)\, du, \tag{10}$$

where $m(t)$ denotes the intensity of the counting process $\{N_d(t), t \geq 0\}$ and f_{σ_M} the density function of σ_M.

The setting of the degradation processes can be considered as a particular case of the framework analysed by Cha and Filkenstein in [44]. In their paper, Cha and Filkenstein considered a system subject to different events arriving according to an NHPP with rate $\nu(t)$. Each event triggers an effective event after random time D_i for $i = 1, 2, \ldots$ with distribution function given by $\lambda(t)$. Denoting by T_e the time to the first effective event, the survival function of T_e at time t is given by

$$P(T_e \geq t) = \exp\left\{ -\int_0^t \nu(u) \lambda(t-u)\, du \right\}. \tag{11}$$

More details about the general case of the effective model are given by [44–46].

Let $S_{[1]}$ be the instant at which, for the first time, the deterioration level of a degradation process exceeds the preventive threshold M,

$$S_{[1]} = \min_{i=1,2,\ldots} \{S_i\}.$$

The probability distribution of $S_{[1]}$ is obtained in the following lemma.

Lemma 2. *Let $S_{[1]}$ be the instant at which, for the first time, the deterioration level of a degradation process exceeds the preventive threshold M. Then, the survival function of $S_{[1]}$ is given by*

$$\bar{F}_{S_{[1]}}(t) = exp\left\{ -\int_0^t m(v) F_{\sigma_M}(t-v)\, dv \right\}, \quad t \geq 0, \tag{12}$$

where $m(t)$ denotes the intensity of $\{N_d(t), t \geq 0\}$ and F_{σ_M} is given by (6).

Equation (12) is obtained by using (11), considering the degradation processes as the successive events and the effective events as the time to reach the preventive threshold M.

The threshold M_s is defined in Section 2.1 as the deterioration level from which the system is more susceptible to a sudden shock. Let σ_{M_s} be the first time at which the deterioration level of the i-th degradation process exceeds the threshold M_s since its initiation for $i = 1, 2, \ldots$. Let V_i be the time where the deterioration level of the i-th degradation process exceeds the threshold M_s. Hence

$$V_i = T_i + \sigma_{M_s}, \quad i = 1, 2, \ldots$$

Let $V_{[1]}$ be the instant at which, for the first time, the deterioration level of a degradation process exceeds the threshold M_s. That is,

$$V_{[1]} = \min_{i=1,2,\ldots} \{V_i\}.$$

Let $N_{M_s}(t)$ be the number of degradation processes whose deterioration level exceed M_s at time t, that is

$$N_{M_s}(t) = \sum_{i=1}^{\infty} \mathbf{1}_{\{V_i \leq t\}}. \tag{13}$$

Using Lemma 1, $\{N_{M_s}(t),\, t \geq 0\}$ is an NHPP with intensity

$$m_{M_s}(t) = \int_0^t m(u) f_{\sigma_{M_s}}(t-u)\, du. \tag{14}$$

By Lemma 2, the survival function of $V_{[1]}$ is given by

$$\bar{F}_{V_{[1]}}(t) = exp\left\{-\int_0^t m(v)F_{\sigma_{M_s}}(t-v)\ dv\right\}, \quad t \geq 0, \tag{15}$$

where $F_{\sigma_{M_s}}$ is given by (6), replacing M by M_s.

2.5. Distribution of sudden shocks

As previously mentioned, the system is also subject to sudden shocks occurring randomly in time. A shock provokes the total failure of the system. Let Y be the time to a sudden shock. Using Equation (3), we can express the stochastic failure rate of Y as

$$h(t) = r_1(t)\mathbf{1}_{\{V_{[1]}>t\}} + r_2(t)\mathbf{1}_{\{V_{[1]}\leq t\}}, \quad t \geq 0.$$

Hence, we have

$$\begin{aligned}\bar{G}(v,t) &= P\left[Y > t|_{V_{[1]}=v}\right] \\ &= \exp\left\{-\int_0^t h(z)\ dz\right\} \\ &= \frac{\bar{F}_1(v)}{\bar{F}_1(0)}\frac{\bar{F}_2(t)}{\bar{F}_2(v)},\end{aligned} \tag{16}$$

for $v \leq t$, where

$$\bar{F}_j(t) = \exp\left\{-\int_0^t r_j(u)\ du\right\}, \quad j = 1, 2, \tag{17}$$

with density functions $f_j(t)$ for $j = 1, 2$.

3. System maintenance

As explained before, the system is inspected each T time units in order to measure the deterioration level of the degradation processes. A preventive replacement is performed when the deterioration level of a degradation process exceeds the preventive threshold M in an inspection time and the system is still working. A corrective replacement is performed when the system is down at an inspection time. After a preventive or a corrective replacement, the system is replaced by a new one. Let R be the replacement cycle length, i.e., the time to a preventive replacement or to a corrective replacement,

$$R = (k+1)T\left[\mathbf{1}_{\{kT<S_{[1]}\leq(k+1)T,(k+1)T<\min\{W_{[1]},Y\}\}} + \mathbf{1}_{\{kT<S_{[1]},kT<\min\{W_{[1]},Y\}\leq(k+1)T\}}\right],$$

for $T > 0$, where $W_{[1]}$ is the instant at which, for the first time, the deterioration level of a degradation process exceeds the failure threshold L. Therefore, the expected time to a replacement cycle is given by

$$E[R] = \sum_{k=1}^{\infty} kT(P_p(kT) + P_c(kT)), \tag{18}$$

where $P_p(kT)$ and $P_c(kT)$ denote the preventive and corrective maintenance probabilities, respectively, at time kT with $k = 1, 2, 3, \ldots$

3.1. Preventive maintenance probability

A preventive replacement is performed at time $(k+1)T$ for $k = 0, 1, 2, \ldots$ if the system is working at time $(k+1)T$ and the threshold M is exceeded for the first time in $(kT, (k+1)T]$. Let $P_p((k+1)T)$ be the preventive maintenance probability at time $(k+1)T$.

After the calculations shown in Appendix A, the probability of a preventive maintenance at time $(k+1)T$ is given by

$$\begin{aligned}P_p((k+1)T) = &P_{p,1}((k+1)T)\mathbf{1}_{\{M\leq M_s\}} \\ &+ P_{p,2}((k+1)T)\mathbf{1}_{\{M>M_s\}},\end{aligned} \tag{19}$$

where

$$\begin{aligned}P_{p,1}((k+1)T) = &\bar{F}_1((k+1)T)\Bigg(\int_{kT}^{(k+1)T} f_{S_{[1]}}(u) \\ &A_{M,M_s}(u,(k+1)T)\ du\Bigg) \\ &+ \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\Bigg(\int_u^{(k+1)T} a_{M,M_s}(u,v) \\ &A_{M_s,L}(v,(k+1)T)\bar{G}(v,(k+1)T)\ dv\Bigg)\ du,\end{aligned}$$

and

$$\begin{aligned}P_{p,2}((k+1)T) = &\int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\Bigg(\bar{G}(v,(k+1)T) \\ &\int_v^{(k+1)T} a_{M_s,M}(v,u)A_{M,L}(u,(k+1)T)\ du\Bigg)\ dv \\ &+ \int_0^{kT} f_{V_{[1]}}(v)\bar{G}(v,(k+1)T)\Bigg(\int_{kT}^{(k+1)T} a_{M_s,M}(v,u) \\ &A_{M,L}(u,(k+1)T)\ du\Bigg)\ dv,\end{aligned}$$

where $f_{S_{[1]}}$ and $f_{V_{[1]}}$ denote the density functions of (12) and (15) and $\bar{G}(u,t)$, $A_{x,y}(u,v)$, and $a_{x,y}(u,v)$ are given by (16), (27), and (28), respectively.

3.2. Corrective maintenance probability

A corrective replacement is performed at time $(k+1)T$ for $k = 0, 1, 2, \ldots$ if the system is down at time $(k+1)T$ and the preventive threshold M is exceeded for the first time after kT. Let $P_c((k+1)T)$ be the corrective maintenance probability at time $(k+1)T$.

After the calculations shown in Appendix B, the probability of a corrective replacement at time $(k+1)T$ is expressed as

$$\begin{aligned}P_c((k+1)T) = &P_{c,1}((k+1)T)\mathbf{1}_{\{M\leq M_s\}} \\ &+ P_{c,2}((k+1)T)\mathbf{1}_{\{M>M_s\}},\end{aligned} \tag{20}$$

where

$$P_{c,1}((k+1)T) = \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\left(\int_{u}^{(k+1)T} a_{M,M_s}(u,v)\int_{v}^{(k+1)T} -\frac{\partial}{\partial w}\left[A_{M_s,L}(v,w)\bar{G}(v,w)\right] dw\ dv\right) du + \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\int_{u}^{(k+1)T} f_1(x)A_{M,M_s}(u,x)\ dx\ du + \int_{kT}^{(k+1)T} f_1(x)\bar{F}_{S_{[1]}}(x)\ dx,$$

and

$$P_{c,2}((k+1)T) = \int_{0}^{(k+1)T} f_{V_{[1]}}(v)\left(\int_{kT}^{(k+1)T} a_{M_s,M}(v,u)\int_{u}^{(k+1)T} -\frac{\partial}{\partial w}\left[A_{M,L}(u,w)\bar{G}(v,w)\right] dw\ du\right) dv + \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\int_{v}^{(k+1)T} a_{M_s,L}(v,w)\bar{G}(v,w)\ dw\ dv + \int_{0}^{kT} f_{V_{[1]}}(v)\int_{kT}^{(k+1)T} g(v,x)A_{M_s,M}(v,x)\ dx\ dv + \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\int_{v}^{(k+1)T} g(v,x)A_{M_s,L}(v,x)\ dx\ dv + \int_{kT}^{(k+1)T} f_1(x)\bar{F}_{V_{[1]}}(x)\ dx,$$

where $f_{S_{[1]}}$ and $f_{V_{[1]}}$ denote the density function of (12) and (15), respectively, $f_1(t)$ is the density function associated with (17) and $\bar{G}(u,t)$, $A_{x,y}(u,v)$, $a_{x,y}(u,v)$, and $g(u,t)$ are given by (16), (27), (28), and (32), respectively.

3.3. Expected downtime

If the system fails, it is down until the next inspection is performed. Let $W_d((k+1)T)$ be the time the system is down in the interval $(kT,(k+1)T]$. Then

$$W_d((k+1)T) = \begin{cases} (k+1)T - Y & \text{if the failure is due to a sudden shock in } (kT,(k+1)T] \\ (k+1)T - W_{[1]} & \text{if the failure is due to degradation in } (kT,(k+1)T] \end{cases}.$$

Based on the calculations of the corrective maintenance probability shown in Appendix B, the expected downtime in $(kT,(k+1)T]$ is given by

$$E[W_d((k+1)T)] = E[W_{d,1}((k+1)T)]\mathbf{1}_{\{M\le M_s\}} + E[W_{d,2}((k+1)T)]\mathbf{1}_{\{M>M_s\}}, \tag{21}$$

being

$$\begin{aligned} &W_{d,1}((k+1)T) \\ &= ((k+1)T-Y)\,\mathbf{1}_{\{kT<S_{[1]}<Y<(k+1)T,Y<W_{[1]}\}} \\ &+ ((k+1)T-Y)\,\mathbf{1}_{\{kT<S_{[1]}<V_{[1]}<Y<(k+1)T,Y<W_{[1]}\}} \\ &+ ((k+1)T-Y)\,\mathbf{1}_{\{kT<Y<(k+1)T,Y<W_{[1]}\}} \\ &+ \left((k+1)T-W_{[1]}\right)\mathbf{1}_{\{kT<S_{[1]}<V_{[1]}<W_{[1]}<(k+1)T,W_{[1]}<Y\}}, \end{aligned}$$

and

$$\begin{aligned} &W_{d,2}((k+1)T) \\ &= ((k+1)T-Y)\,\mathbf{1}_{\{V_{[1]}<kT<S_{[1]}<Y<(k+1)T,Y<W_{[1]}\}} \\ &+ ((k+1)T-Y)\,\mathbf{1}_{\{V_{[1]}<kT<Y<(k+1)T,Y<S_{[1]}\}} \\ &+ ((k+1)T-Y)\,\mathbf{1}_{\{kT<V_{[1]}<Y<(k+1)T,Y<W_{[1]}\}} \\ &+ ((k+1)T-Y)\,\mathbf{1}_{\{kT<Y<(k+1)T,Y<V_{[1]}\}} \\ &+ \left((k+1)T-W_{[1]}\right)\mathbf{1}_{\{V_{[1]}<kT<S_{[1]}<W_{[1]}<(k+1)T,W_{[1]}<Y\}} \\ &+ \left((k+1)T-W_{[1]}\right)\mathbf{1}_{\{kT<V_{[1]}<W_{[1]}<(k+1)T,W_{[1]}<Y\}}. \end{aligned}$$

That is

$$E[W_{d,1}((k+1)T)] = \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\left(\int_{u}^{(k+1)T} f_1(x) A_{M,M_s}(u,x)((k+1)T-x)\ dx\right) du + \int_{kT}^{(k+1)T} f_1(x)\bar{F}_{S_{[1]}}(x)((k+1)T-x)\ dx + \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\left(\int_{u}^{(k+1)T} a_{M,M_s}(u,v)\int_{v}^{(k+1)T} -\frac{\partial}{\partial w}\left[A_{M_s,L}(v,w)\bar{G}(v,w)\right]((k+1)T-w)\ dw\ dv\right) du,$$

and

$$E[W_{d,2}((k+1)T)] = \int_{0}^{kT} f_{V_{[1]}}(v)\left(\int_{kT}^{(k+1)T} g(v,x) A_{M_s,M}(v,x)((k+1)T-x)\ dx\right) dv + \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\left(\int_{v}^{(k+1)T} g(v,x)A_{M_s,L}(v,x) ((k+1)T-x)\ dx\right) dv$$

$$
\begin{aligned}
&+\int_{kT}^{(k+1)T} f_1(x)\bar{F}_{V_{[1]}}(x)((k+1)T-x)\ dx \\
&+\int_{0}^{(k+1)T} f_{V_{[1]}}(v)\Bigg(\int_{kT}^{(k+1)T} a_{M_s,M}(v,u) \\
&\int_{u}^{(k+1)T} -\frac{\partial}{\partial w}\left[A_{M,L}(u,w)\bar{G}(v,w)\right] \\
&((k+1)T-w)\ dw\ du\Bigg)\ dv \\
&+\int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\Bigg(\int_{v}^{(k+1)T} a_{M_s,L}(v,w)\bar{G}(v,w) \\
&((k+1)T-w)\ dw\Bigg)\ dv
\end{aligned}
$$

3.4. Expected number of inspections

Let $N_i(R)$ be the total number of inspections performed on the system in a replacement cycle, then

$$
\begin{aligned}
E[N_i(R)] &= \frac{E[R]}{T} \\
&= \sum_{k=1}^{\infty} k(P_c(kT)+P_p(kT)), \qquad (22)
\end{aligned}
$$

where $P_p(kT)$ and $P_c(kT)$ are the probabilities of a preventive maintenance and a corrective maintenance at time kT for $k=1,2\ldots$ given by (19) and (20), respectively.

3.5. Expected cost rate

An aim of this paper is to assess an optimal maintenance strategy. By optimal, we mean the values T and M which minimize the objective cost function given by (1). Let $C(T,M)$ be the expected cost rate for this maintenance model. Hence,

$$
\begin{aligned}
C(T,M) = \Bigg(\sum_{k=0}^{\infty}\Bigg(&C_cP_c((k+1)T)+C_pP_p((k+1)T) \\
&+C_dE[W_d((k+1)T)]\Bigg)+C_iE[N_i(R)]\Bigg)\Bigg/E[R],
\end{aligned} \qquad (23)
$$

where $P_p((k+1)T)$, $P_c((k+1)T)$, $E[W_d(k+1(T))]$, $E[N_i(R)]$, and $E[R]$ are given by (19), (20), (21), (22), and (18), respectively. The optimization problem is reduced to find the values T_{opt} and M_{opt} such that

$$
C(T_{opt},M_{opt}) = \inf\left\{C(T,M),\ T>0,\ 0<M\leq L\right\},
$$

where $C(T,M)$ is given by (23). Due to the analytical complexity of $C(T,M)$, the optimization of the expected cost rate $C(T,M)$ for a data set is performed in the next section using numerical methods.

4. Illustrative example

We assume a system subject to a pitting corrosion process where the initiation of degradation processes is distributed according to an NHPP with power-law intensity

$$
m(t) = \frac{b}{a^b}t^{b-1}, \qquad t\geq 0,
$$

where a and b ($a,b>0$) are the scale and shape parameters of the process, respectively. This model is one of the most widely used to analyse failure data from reliability growth studies (see, e.g. [47] and [48]). For this example, we consider $a=7$ and $b=5$.

The growth of degradation processes is modelled according to a gamma process with parameters $\alpha=\beta=0.2$. Let $X_i(t)$ be the deterioration level of the i-th degradation process t time units after its initiation ($i=1,2,\ldots$). The density function of $X_i(t)$ is given by

$$
f_{0.2t,0.2}(x) = \frac{0.2^{0.2t}}{\Gamma(0.2t)}x^{0.2t-1}e^{-0.2x}, \quad x\geq 0.
$$

The system fails due to both degradation and sudden shocks. A degradation failure occurs when the deterioration level of an existing degradation process in the system exceeds the deterioration threshold $L=20$. A dependence model is considered between the two competing causes of failure. This dependence is reflected in the fact in the failure rate function of the sudden shock process. That is, the system is more susceptible to an sudden shock when the deterioration level of a degradation process exceeds the level M_s. We assume that $M_s=15$ and that the sudden shock process follows an NHPP with intensity

$$
\begin{aligned}
h(t) = 0.01 &\prod_{i=1}^{N_d(t)} \mathbf{1}_{\{X_i^*(t)\leq 15\}} \\
&+0.10\left[1-\prod_{i=1}^{N_d(t)} \mathbf{1}_{\{X_i^*(t)\leq 15\}}\right], \quad t\geq 0,
\end{aligned}
$$

where $X_i^*(t)$ denotes the deterioration level of the i-th degradation process at time t and $N_d(t)$ is the number of degradation processes in the system at time t. Under these specifications, the expected time until a degradation failure is 18.96 $t.u.$ (time units) and the expected time to a sudden shock is 24.58 $t.u.$.

We assume that the sequence of costs is $C_i=7$ $m.u.$ (monetary units), $C_d=46$ $m.u.$, $C_p=281$ $m.u.$ and $C_c=442$ $m.u.$ Then, the expected cost rate as a function of the deterioration level M and the time between inspections T is shown in Figure 2. The convexity of the cost surface indicates the existence of an optimal setting of parameters T and M which minimize the expected cost rate.

A solution has been numerically found by simulation. The simulation model has been implemented in MATLAB software. A grid of size 50 in the interval $(0,10)$ has been considered for T. Similarly, a grid of size 50 in the interval

$(0, 20)$ has been considered for M. For each combination of M and T, 25000 iterations were performed. As a result, $T_{opt} = 3.60$ $t.u.$ and $M_{opt} = 12.40$ are obtained. The minimal expected cost rate is $C(T_{opt}, M_{opt}) = 27.13$ $m.u/t.u.$

Now, we focus on the influence of the main model parameters on the solution. Firstly, a sensitivity analysis of the gamma process parameter is performed. Later, the power-law intensity parameters are considered.

The values of the gamma process parameters are modified according to the following specifications:

$$\alpha_{(v_i\%)} = \alpha \left[1 + \frac{v_i}{100}\right] \quad \text{and} \quad \beta_{(v_j\%)} = \beta \left[1 + \frac{v_j}{100}\right],$$

where v_i and v_j are, respectively, the i-th and j-th position of the vector $\mathbf{v} = (-10, -5, -1, 0, 1, 5, 10)$. Then, the parameter values for α and β can be simultaneous and independently modified both for increasing and decreasing changes.

Let $C_{\alpha_{(v_i\%)},\beta_{(v_j\%)}}$ be the minimal expected cost rate obtained by varying the gamma process parameters simultaneously. Then, a relative measure is defined as

$$V_{\alpha_{(v_i\%)},\beta_{(v_j\%)}} = \frac{|C(T_{opt}, M_{opt}) - C_{\alpha_{(v_i\%)},\beta_{(v_j\%)}}|}{C(T_{opt}, M_{opt})},$$

where $C(T_{opt}, M_{opt})$ is the minimal expected cost rate previously calculated with the original parameter values.

For fixed i and j, $V_{\alpha_{(v_i\%)},\beta_{(v_j\%)}}$ measures the relative difference between the current optimal cost and the optimal cost that has been calculated by using the modified parameter values. If this quantity is multiplied by 100, the result is expressed in percentage. The closer to zero, the less influence on the solution the modified parameter values have.

Table 1 shows the relative variation percentages with a shaded grey scale. Each cell represents $V_{\alpha_{(v_i\%)},\beta_{(v_j\%)}}$ multiplied by 100. Darker colours of cells denote a higher relative variation percentage. By modifying $\pm 5\%$ around $\alpha = 0.2$ and $\beta = 0.2$, the relative variation percentages are small. The results also show that the relative variation percentages are lower in the diagonal of the table. That is, when the parameters α and β are modified in the same direction and magnitude.

Similarly, the values of the power-law intensity parameters are modified according to the following specifications:

$$a_{(v_i\%)} = a \left[1 + \frac{v_i}{100}\right] \quad \text{and} \quad b_{(v_j\%)} = b \left[1 + \frac{v_j}{100}\right].$$

Let $C_{a_{(v_i\%)},b_{(v_j\%)}}$ be the minimal expected cost rate obtained by varying the power-law intensity parameters simultaneously as in the previous scheme. Now, the relative measure is given by

$$V_{a_{(v_i\%)},b_{(v_j\%)}} = \frac{|C(T_{opt}, M_{opt}) - C_{a_{(v_i\%)},b_{(v_j\%)}}|}{C(T_{opt}, M_{opt})}.$$

The relative variation percentages are presented in Table 2. Again, by modifying $\pm 5\%$ around $a = 7$ and $b = 5$, the relative variation percentages are small. The results show that the parameter a has greater effects on $V_{a_{(v_i\%)},b_{(v_j\%)}}$ than the parameter b.

5. Conclusion and future extensions

A condition-based maintenance of a system subject to both multiple degradation processes and sudden shocks is considered in this paper. In order to complete the study of the maintenance strategy, a numerical search of the optimal values is performed and the robustness of the solution when varying different parameters, is analysed.

In this paper, the maintenance strategy is studied assuming the dependence between the competing causes of failure. This dependence is analysed under the approach that the intensity of the process of sudden shocks increases with the deterioration level of the degradation processes. Some further possible extension of this model are to consider that the threshold M_s could be random and dependent of other factors and to analyse other models of dependence between the causes of failure. For example, considering that the intensity of the sudden shocks depends, not only on the deterioration level of the degradation processes, but also on the number of degradation process existing in the system.

Appendix A: Preventive maintenance probability calculation

The preventive maintenance probability $P_p((k+1)T)$ for $k = 0, 1, 2, \ldots$ is expressed as

$$\begin{aligned} P_p((k+1)T) = P_{p,1}((k+1)T)\mathbf{1}_{\{M \le M_s\}} \\ + P_{p,2}((k+1)T)\mathbf{1}_{\{M > M_s\}}, \end{aligned} \tag{24}$$

where $P_{p,1}((k+1)T)$ denotes the preventive maintenance probability at time $(k+1)T$ if $M \le M_s$ and $P_{p,2}((k+1)T)$ denotes the preventive maintenance probability at time $(k+1)T$ if $M > M_s$.

Firstly, we assume $M \le M_s$. A preventive replacement is performed at time $(k+1)T$ if one of the following mutually exclusive events take place

$$\left\{kT < S_{[1]} < (k+1)T < V_{[1]}, Y > (k+1)T\right\},$$

$$\left\{kT < S_{[1]} < V_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T\right\}.$$

Hence,

$$P_{p,1}((k+1)T) = P_{p,1,1}((k+1)T) + P_{p,1,2}((k+1)T),$$

where

$$\begin{aligned} &P_{p,1,1}((k+1)T) \\ &= P\left[kT < S_{[1]} < (k+1)T < V_{[1]}, Y > (k+1)T\right], \end{aligned}$$

and

$$P_{p,1,2}((k+1)T) \\ = P\left[kT < S_{[1]} < V_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T\right].$$

Hence,

$$P_{p,1,1}((k+1)T) = \int_{kT}^{(k+1)T} f_{S_{[1]}}(u) \\ \bar{F}_{\sigma_{M_s}-\sigma_M}((k+1)T-u) \sum_{i=0}^{\infty} P\left[(k+1)T < V_{[1]}, \right. \\ \left. N_M(u,(k+1)T) = i, Y > (k+1)T\right] \, du.$$

On the other hand

$$\sum_{i=0}^{\infty} P[(k+1)T < V_{[1]}, N_M(u,(k+1)T) = i] \\ = \exp\left\{-\int_u^{(k+1)T} m_M(z)dz\right\} \\ \left(1 + \sum_{i=1}^{\infty} B_{u_1,i}((k+1)T)\right),$$

being

$$B_{u_1,i}(t) = \int_{u_1}^{t} b_t(u_2)\, du_2 \int_{u_2}^{t} b_t(u_3)\, du_3 \ldots \\ \int_{u_{i-1}}^{t} b_t(u_i)\, du_i \int_{u_i}^{t} b_t(u_{i+1})\, du_{i+1}, \tag{25}$$

where

$$b_t(u) = m_M(u)\bar{F}_{\sigma_{M_s}-\sigma_M}(t-u), \quad u \le t.$$

Let $Z_{M,M_s}(u_1,t)$ be the function given by

$$Z_{M,M_s}(u_1,t) \\ = \exp\left\{-\int_{u_1}^{t} m_M(z)dz\right\}\left(1 + \sum_{i=1}^{\infty} B_{u_1,i}(t)\right),$$

where $B_{u_1,i}(t)$ is given by (25). Using Lemma C.1 given in [33], $Z_{M,M_s}(u_1,t)$ can be simplified as

$$Z_{M,M_s}(u_1,t) = \exp\left\{-\int_{u_1}^{t} m_M(z)F_{\sigma_{M_s}-\sigma_M}(t-z)dz\right\}. \tag{26}$$

Hence,

$$P_{p,1,1}((k+1)T) = \bar{F}_1((k+1)T)\left(\int_{kT}^{(k+1)T} f_{S_{[1]}}(u) \right. \\ \left. \bar{F}_{\sigma_{M_s}-\sigma_M}((k+1)T-u) Z_{M,M_s}(u,(k+1)T)\, du\right).$$

Let $A_{M,M_s}(u,v)$ be the function which represents the probability that the deterioration level M_s is not exceeded at time v by a degradation process, given that the preventive threshold M was exceeded for the first time at u. That is,

$$A_{M,M_s}(u,v) = P\left[V_{[1]} > v|_{S_{[1]}=u}\right] \\ = \bar{F}_{\sigma_{M_s}-\sigma_M}(v-u) Z_{M,M_s}(u,v), \tag{27}$$

where $Z_{M,M_s}(u,v)$ is given by (26).

Let $a_{M,M_s}(u,v)$ be derivative function of (27). That is,

$$a_{M,M_s}(u,v) = \frac{-\partial}{\partial v} A_{M,M_s}(u,v). \tag{28}$$

Hence,

$$P_{p,1,2}((k+1)T) = \int_{kT}^{(k+1)T} f_{S_{[1]}}(u) P\left[u < V_{[1]} < \right. \\ \left. (k+1)T < W_{[1]}, Y > (k+1)T|_{S_{[1]}=u}\right] \, du \\ = \int_{kT}^{(k+1)T} f_{S_{[1]}}(u) \int_u^{(k+1)T} a_{M,M_s}(u,v) \\ P\left[(k+1)T < W_{[1]}, Y > (k+1)T|_{V_{[1]}=v}\right] \, du\, dv \\ = \int_{kT}^{(k+1)T} f_{S_{[1]}}(u) \int_u^{(k+1)T} a_{M,M_s}(u,v) \\ A_{M_s,L}(v,(k+1)T) P\left[Y > (k+1)T|_{V_{[1]}=v}\right] \, du\, dv \\ = \int_{kT}^{(k+1)T} f_{S_{[1]}}(u) \int_u^{(k+1)T} a_{M,M_s}(u,v) \\ A_{M_s,L}(v,(k+1)T)\bar{G}(v,(k+1)T)\, du\, dv$$

where $f_{S_{[1]}}$ denotes the density function of (12) and $\bar{G}(u,t)$, $A_{M_s,L}(v,(k+1)T)$, and $a_{M,M_s}(u,v)$ are given by (16), (27), and (28), respectively.

Therefore, the preventive maintenance probability for $M \le M_s$ is given by

$$P_{p,1}((k+1)T) = \bar{F}_1((k+1)T)\left(\int_{kT}^{(k+1)T} f_{S_{[1]}}(u) \right. \\ \left. A_{M,M_s}(u,(k+1)T)\, du\right) \\ + \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\left(\int_u^{(k+1)T} a_{M,M_s}(u,v) \right. \\ \left. A_{M_s,L}(v,(k+1)T)\bar{G}(v,(k+1)T)\, dv\right) du. \tag{29}$$

Considering $M > M_s$, a preventive replacement is performed at time $(k+1)T$ if one of the following mutually exclusive events occurs

$$\left\{kT < V_{[1]} < S_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T\right\},$$

$$\left\{V_{[1]} < kT < S_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T\right\}.$$

Hence,

$$P_{p,2}((k+1)T) = P_{p,2,1}((k+1)T) + P_{p,2,2}((k+1)T),$$

where

$$P_{p,2,1}((k+1)T) = P\left[kT < V_{[1]} < S_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T\right],$$

and $P_{p,2,2}((k+1)T)$ is

$$P_{p,2,2}((k+1)T) = P\left[V_{[1]} < kT < S_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T\right].$$

Hence,

$$\begin{aligned}
P_{p,2,1}((k+1)T) &= \int_{kT}^{(k+1)T} f_{V_{[1]}}(v) P\left[v < S_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T |_{V_{[1]}=v}\right] dv \\
&= \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\bar{G}(v,(k+1)T)\Bigg(P\left[v < S_{[1]} < (k+1)T < W_{[1]}\right]\Bigg) dv \\
&= \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\bar{G}(v,(k+1)T)\Bigg(\int_{v}^{(k+1)T} a_{M_s,M}(v,u) P\left[(k+1)T < W_{[1]}|_{S_{[1]}=u}\right] du\Bigg) dv \\
&= \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\bar{G}(v,(k+1)T)\Bigg(\int_{v}^{(k+1)T} a_{M_s,M}(v,u) A_{M,L}(u,(k+1)T)\, du\Bigg) dv,
\end{aligned}$$

and

$$\begin{aligned}
P_{p,2,2}((k+1)T) &= \int_{0}^{kT} f_{V_{[1]}}(v) P\left[kT < S_{[1]} < (k+1)T < W_{[1]}, Y > (k+1)T |_{V_{[1]}=v}\right] dv \\
&= \int_{0}^{kT} f_{V_{[1]}}(v)\bar{G}(v,(k+1)T)\Bigg(P\left[kT < S_{[1]} < (k+1)T < W_{[1]}|_{V_{[1]}=v}\right]\Bigg) dv \\
&= \int_{0}^{kT} f_{V_{[1]}}(v)\bar{G}(v,(k+1)T)\Bigg(\int_{kT}^{(k+1)T} a_{M_s,M}(v,u) P\left[(k+1)T < W_{[1]}|_{S_{[1]}=u}\right] du\Bigg) dv \\
&= \int_{0}^{kT} f_{V_{[1]}}(v)\bar{G}(v,(k+1)T)\Bigg(\int_{kT}^{(k+1)T} a_{M_s,M}(v,u) A_{M,L}(u,(k+1)T)\, du\Bigg) dv.
\end{aligned}$$

Therefore, the expression for the preventive maintenance probability when $M > M_s$ is given by

$$\begin{aligned}
P_{p,2}((k+1)T) &= \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\Bigg(\int_{v}^{(k+1)T} a_{M_s,M}(v,u) A_{M,L}(u,(k+1)T)\bar{G}(v,(k+1)T)\, du\Bigg) dv \\
&+ \int_{0}^{kT} f_{V_{[1]}}(v)\Bigg(\int_{kT}^{(k+1)T} a_{M_s,M}(v,u) A_{M,L}(u,(k+1)T)\bar{G}(v,(k+1)T)\, du\Bigg) dv,
\end{aligned} \quad (30)$$

where $f_{V_{[1]}}$ denotes the density function of (15) and $\bar{G}(u,t)$, $A_{M,L}(u,v)$, and $a_{M_s,M}(v,u)$ are given by (16), (27), and (28), respectively.

Appendix B: Corrective maintenance probability

The corrective maintenance probability $P_c((k+1)T)$ at time $(k+1)T$ for $k = 0,1,2,\dots$ is expressed as

$$\begin{aligned} P_c((k+1)T) = P_{c,1}((k+1)T)\mathbf{1}_{\{M \le M_s\}} \\ + P_{c,2}((k+1)T)\mathbf{1}_{\{M > M_s\}}, \end{aligned} \quad (31)$$

where $P_{c,1}((k+1)T)$ denotes the corrective maintenance probability at time $(k+1)T$ if $M \le M_s$ and $P_{c,2}((k+1)T)$ denotes the corrective maintenance probability at time $(k+1)T$ if $M > M_s$.

In the interval $(kT,(k+1)T]$, a corrective replacement is performed if the failure of the system is provoked when a deterioration level of a degradation process exceeds the failure threshold L, as well as a sudden shock arrives at the system in $(kT,(k+1)T]$.

For $M \le M_s$, the corrective replacement due to degradation occurs if

$$\left\{kT < S_{[1]} < V_{[1]} < W_{[1]} < (k+1)T, W_{[1]} < Y\right\},$$

and a corrective replacement due to a sudden shock is given by the occurrence of one of the following mutually exclusive events

$$\left\{kT < S_{[1]} < Y < (k+1)T, Y < V_{[1]}\right\},$$
$$\left\{kT < S_{[1]} < V_{[1]} < Y < (k+1)T, Y < W_{[1]}\right\},$$
$$\left\{kT < Y < (k+1)T, Y < S_{[1]}\right\}.$$

Hence,

$$P_{c,1}((k+1)T) = \sum_{i=1}^{4} P_{c,1,i}((k+1)T),$$

where the probabilities $P_{c,1,i}((k+1)T)$ for $i = 1,2,3,4$ are given by

$$\begin{aligned} &P_{c,1,1}((k+1)T) \\ &= P\left[kT < S_{[1]} < V_{[1]} < W_{[1]} < (k+1)T, W_{[1]} < Y\right], \end{aligned}$$

$$P_{c,1,2}((k{+}1)T) = P\left[kT < S_{[1]} < Y < (k+1)T, Y < V_{[1]}\right],$$

$$P_{c,1,3}((k+1)T) = P\left[kT < S_{[1]} < V_{[1]} < Y < (k+1)T, Y < W_{[1]}\right],$$

and

$$P_{c,1,4}((k+1)T) = P\left[kT < Y < (k+1)T, Y < S_{[1]}\right],$$

respectively.

Let $g(u,t)$ be the function

$$g(u,t) = \frac{-\partial}{\partial t}\bar{G}(u,t), \quad u \le t, \tag{32}$$

where $\bar{G}(u,t)$ is given by (16).

Following the development of the preventive maintenance probability calculation, the corrective maintenance probability at time $(k+1)T$ when $M \le M_s$ is given by

$$\begin{aligned}
P_{c,1}((k+1)T) &= \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\left(\int_{u}^{(k+1)T} a_{M,M_s}(u,v)\right.\\
&\quad \left.\int_{v}^{(k+1)T} -\frac{\partial}{\partial w}\left[A_{M_s,L}(v,w)\bar{G}(v,w)\right]\, dw\ dv\right) du\\
&\quad + \int_{kT}^{(k+1)T} f_{S_{[1]}}(u)\int_{u}^{(k+1)T} f_1(x)A_{M,M_s}(u,x)\ dx\ du\\
&\quad + \int_{kT}^{(k+1)T} f_1(x)\bar{F}_{S_{[1]}}(x)\ dx.
\end{aligned}$$

For $M > M_s$, a corrective replacement at time $(k+1)T$ due to degradation is performed if one of the following mutually exclusive events takes place

$$\left\{V_{[1]} < kT < S_{[1]} < W_{[1]} < (k+1)T, Y > W_{[1]}\right\},$$

$$\left\{kT < V_{[1]} < W_{[1]} < (k+1)T, Y > W_{[1]}\right\},$$

and due to a sudden shock

$$\left\{V_{[1]} < kT < S_{[1]} < Y < (k+1)T, Y < W_{[1]}\right\},$$

$$\left\{V_{[1]} < kT < Y < (k+1)T, Y < S_{[1]}\right\},$$

$$\left\{kT < V_{[1]} < Y < (k+1)T, Y < W_{[1]}\right\},$$

$$\left\{kT < Y < (k+1)T, Y < V_{[1]}\right\}.$$

Hence,

$$P_{c,2}((k+1)T) = \sum_{i=1}^{6} P_{c,2,i}((k+1)T),$$

where $P_{c,2,i}((k+1)T)$ for $i = 1, 2, \ldots, 6$ are given by

$$P_{c,2,1}((k+1)T) = P\left[V_{[1]} < kT < S_{[1]} < W_{[1]} < (k+1)T, Y > W_{[1]}\right],$$

$$P_{c,2,2}((k+1)T) = P\left[kT < V_{[1]} < W_{[1]} < (k+1)T, Y > W_{[1]}\right],$$

$$P_{c,2,3}((k+1)T) = P\left[V_{[1]} < kT < S_{[1]} < Y < (k+1)T, Y < W_{[1]}\right],$$

$$P_{c,2,4}((k{+}1)T) = P\left[V_{[1]} < kT < Y < (k+1)T, Y < S_{[1]}\right],$$

$$P_{c,2,5}((k{+}1)T) = P\left[kT < V_{[1]} < Y < (k+1)T, Y < W_{[1]}\right],$$

and

$$P_{c,2,6}((k+1)T) = P\left[kT < Y < (k+1)T, Y < V_{[1]}\right],$$

respectively.

Thus, based on previous calculations, $P_{c,2}((k+1)T)$ is given by

$$\begin{aligned}
P_{c,2}((k+1)T) &= \int_{0}^{(k+1)T} f_{V_{[1]}}(v)\left(\int_{kT}^{(k+1)T} a_{M_s,M}(v,u)\right.\\
&\quad \left.\int_{u}^{(k+1)T} -\frac{\partial}{\partial w}\left[A_{M,L}(u,w)\bar{G}(v,w)\right]\, dw\ du\right) dv\\
&\quad + \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\int_{v}^{(k+1)T} a_{M_s,L}(v,w)\bar{G}(v,w)\ dw\ dv\\
&\quad + \int_{0}^{kT} f_{V_{[1]}}(v)\int_{kT}^{(k+1)T} g(v,x)A_{M_s,M}(v,x)\ dx\ dv\\
&\quad + \int_{kT}^{(k+1)T} f_{V_{[1]}}(v)\int_{v}^{(k+1)T} g(v,x)A_{M_s,L}(v,x)\ dx\ dv\\
&\quad + \int_{kT}^{(k+1)T} f_1(x)\bar{F}_{V_{[1]}}(x)\ dx.
\end{aligned}$$

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[1] I. Alsyouf, The role of maintenance in improving companies' productivity and profitability, International Journal of Production Economics 105 (1) (2007) 70–78.

[2] R. Barlow, L. Hunter, Optimum preventive maintenance policies, Operations Research 8 (1) (1960) 90–100.

[3] R. Radner, D. W. Jorgenson, Optimal replacement and inspection of stochastically failing equipment, Stanford University Press, Stanford, California, 1962.

[4] J. J. McCall, Operating characteristics of opportunistic replacement and inspection policies, Management Science 10 (1) (1963) 85–97.

[5] M. Rausand, A. Høyland, System Reliability Theory: Models, Statistical Methods, and Applications, Wiley series in probability and statistics: Applied probability and statistics, Wiley-Interscience, 2004.

[6] X. Zhao, M. Fouladirad, C. Bérenguer, L. Bordes, Condition-based inspection/replacement policies for non-monotone deteriorating systems with environmental covariates, Reliability Engineering & System Safety 95 (8) (2010) 921–934.

[7] M. Fouladirad, A. Grall, Condition-based maintenance for a system subject to a non-homogeneous wear process with a wear rate transition, Reliability Engineering & System Safety 96 (6) (2011) 611–618.
[8] R. Jiang, A multivariate CBM model with a random and time-dependent failure threshold, Reliability Engineering & System Safety 119 (0) (2013) 178–185.
[9] H. P. Hong, W. Zhou, S. Zhang, W. Ye, Optimal condition-based maintenance decisions for systems with dependent stochastic degradation of components, Reliability Engineering & System Safety 121 (2014) 276–288.
[10] S. D. Cramer, B. S. Covino, ASM Handbook vol. 13 A Corrosion: Fundamentals, Testing, and Protection, Materials Park, OH: ASM International, 2003.
[11] R. Salvarezza, N. de Cristofaro, C. Pallotta, A. Arvia, Stochastic and deterministic behaviours of 316 stainless steel pitting corrosion in phosphate-borate buffer containing sodium chloride, Electrochimica Acta 32 (7) (1987) 1049–1055.
[12] D. Williams, C. Westcott, M. Fleischmann, Studies of the initiation of pitting corrosion on stainless steels, Journal of Electroanalytical Chemistry and Interfacial Electrochemistry 180 (12) (1984) 549–564.
[13] A. Valor, F. Caleyo, L. Alfonso, D. Rivas, J. Hallen, Stochastic modeling of pitting corrosion: A new model for initiation and growth of multiple corrosion pits, Corrosion Science 49 (2) (2007) 559–579.
[14] A. Sheikh, J. Boah, D. Hansen, Statistical modeling of pitting corrosion and pipeline reliability, Corrosion 46 (3) (1990) 190–197.
[15] J. van Noortwijk, H. Klatter, Optimal inspection decisions for the block mats of the Eastern-Scheldt barrier, Reliability Engineering & System Safety 65 (3) (1999) 203–211.
[16] S. Mercier, L. Bordes, E. Remy, E. Dautrême, A stochastic model for competing degradations, Safety, Reliability and Risk Analysis: Beyond the Horizon (2014) 1731–1738.
[17] S. P. Kuniewski, J. A. van der Weide, J. M. van Noortwijk, Sampling inspection for the evaluation of time-dependent reliability of deteriorating systems under imperfect defect detection, Reliability Engineering & System Safety 94 (9) (2009) 1480–1490.
[18] I. T. Castro, A. Barros, A. Grall, Age-based preventive maintenance for passive components submitted to stress corrosion cracking, Mathematical and Computer Modelling 54 (1-2) (2011) 598–609.
[19] M. Abdel-Hameed, A gamma wear process, Reliability, IEEE Transactions on R-24 (2) (1975) 152–153.
[20] J. van Noortwijk, A survey of the application of gamma processes in maintenance, Reliability Engineering & System Safety 94 (1) (2009) 2–21.
[21] A. J. Lemoine, M. L. Wenocur, On failure modeling, Naval Research Logistics 32 (3) (1985) 497–508.
[22] Y. H. Lin, Y. F. Li, E. Zio, Multi-state physics model for the reliability assessment of a component under degradation processes and random shocks, Safety, Reliability and Risk Analysis: Beyond the Horizon (2014) 315–321.
[23] Z. Wang, H.-Z. Huang, Y. Li, N.-C. Xiao, An approach to reliability assessment under degradation and shock process, Reliability, IEEE Transactions on 60 (4) (2011) 852–863.
[24] N. D. Singpurwalla, Survival in dynamic environments, Statistical Science 10 (1) (1995) 86–103.
[25] D. Cox, Some remarks on failure-times, surrogate markers, degradation, wear, and the quality of life, Lifetime Data Analysis 5 (1999) 307–314.
[26] W. Li, H. Pham, A condition-based inspection-maintenance model based on geometric sequences for systems with a degradation process and random shocks, Life Cycle Reliability and Safety Engineering 1 (1) (2012) 26–34.
[27] K. T. Huynh, I. T. Castro, A. Barros, C. Bérenguer, Modeling age-based maintenance strategies with minimal repairs for systems subject to competing failure modes due to degradation and shocks, European Journal of Operational Research 218 (1) (2012) 140–151.
[28] K. T. Huynh, A. Barros, C. Bérenguer, I. T. Castro, A periodic inspection and replacement policy for systems subject to competing failure modes due to degradation and traumatic events, Reliability Engineering & System Safety 96 (4) (2011) 497–508.
[29] S. M. Asadzadeh, A. Azadeh, An integrated systemic model for optimization of condition-based maintenance with human error, Reliability Engineering & System Safety 124 (2014) 117–131.
[30] L. XiaoFei, L. Min, Hazard rate function in dynamic environment, Reliability Engineering & System Safety 130 (2014) 50–60.
[31] E. Deloux, B. Castanier, C. Bérenguer, Predictive maintenance policy for a gradually deteriorating system subject to stress, Reliability Engineering & System Safety 94 (2) (2009) 418–431.
[32] I. T. Castro, An age-based maintenance strategy for a degradation-threshold-shock-model for a system subjected to multiple defects, Asia-Pacific Journal of Operational Research 30 (05) (2013) 1350016–1350029.
[33] I. T. Castro, N. C. Caballé, C. J. Pérez, A condition-based maintenance for a system subject to multiple degradation processes and external shocks, International Journal of Systems Science 0 (0) (0) 1–13. doi:10.1080/00207721.2013.828796.
[34] W. Li, H. Pham, Reliability modeling of multi-state degraded systems with multi-competing failures and random shocks, Reliability, IEEE Transactions on 54 (2) (2005) 297–303.
[35] Y. Wang, H. Pham, Modeling the dependent competing risks with multiple degradation processes and random shock using time-varying copulas, Reliability, IEEE Transactions on 61 (1) (2012) 13–22.
[36] D. R. Cox, Some statistical methods connected with series of events, Journal of the Royal Statistical Society. Series B (Methodological) (1955) 129–164.
[37] D. Cox, P. Lewis, The Statistical Analysis of Series of Events, Monographs on Statistics and Applied Probability, Chapman and Hall, 1966.
[38] W. Kahle, H. Wendt, On a cumulative damage process and resulting first passages times, Applied Stochastic Models in Business and Industry 20 (1) (2004) 17–26.
[39] I. Castro, A model of imperfect preventive maintenance with dependent failure modes, European Journal of Operational Research 196 (1) (2009) 217–224.
[40] H. Tijms, A First Course in Stochastic Models, John Wiley and Sons Inc., 2003.
[41] M. Finkelstein, Failure Rate Modelling for Reliability and Risk, Springer, 2008.
[42] R. Nicolai, Maintenance models for systems subject to measurable deterioration, Ph.D. thesis, Erasmus University Rotterdam, Rotterdam, The Netherlands (March 2008).
[43] J. Bertoin, Lévy Processes, Cambridge Tracts in Mathematics, Cambridge University Press, 1998.
[44] J. H. Cha, M. Finkelstein, A note on 'curable' shock processes, Journal of Statistical Planning and Inference 142 (12) (2012) 3146–3151.
[45] J. H. Cha, M. Finkelstein, Stochastic survival models with events triggered by external shocks, Probability in the Engineering and Informational Sciences 26 (02) (2012) 183–195.
[46] M. Finkelstein, J. H. Cha, Stochastic Modelling for Reliability: Shocks Burn-in, and Heterogeneous Populations, Springer, 2013.
[47] M. Guida, R. Calabria, G. Pulcini, Bayes inference for a non-homogeneous Poisson process with power intensity law, Reliability, IEEE Transactions on 38 (5) (1989) 603–609.
[48] R. Calabria, G. Pulcini, Bayes inference for the modulated power law process, Communications in Statistics - Theory and Methods 26 (10) (1997) 2421–2438.

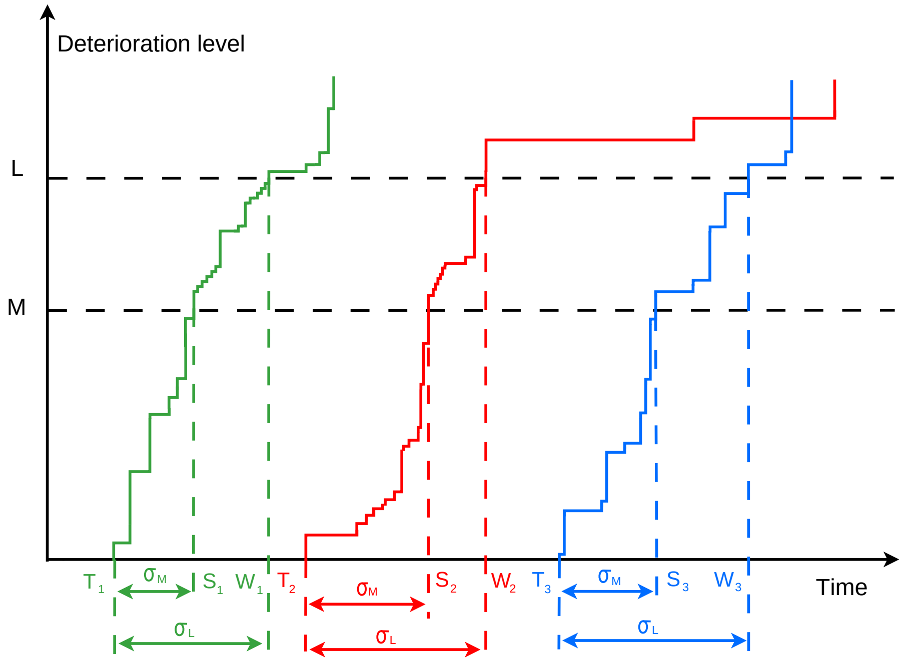

Fig. 1: Realization of the degradation processes.

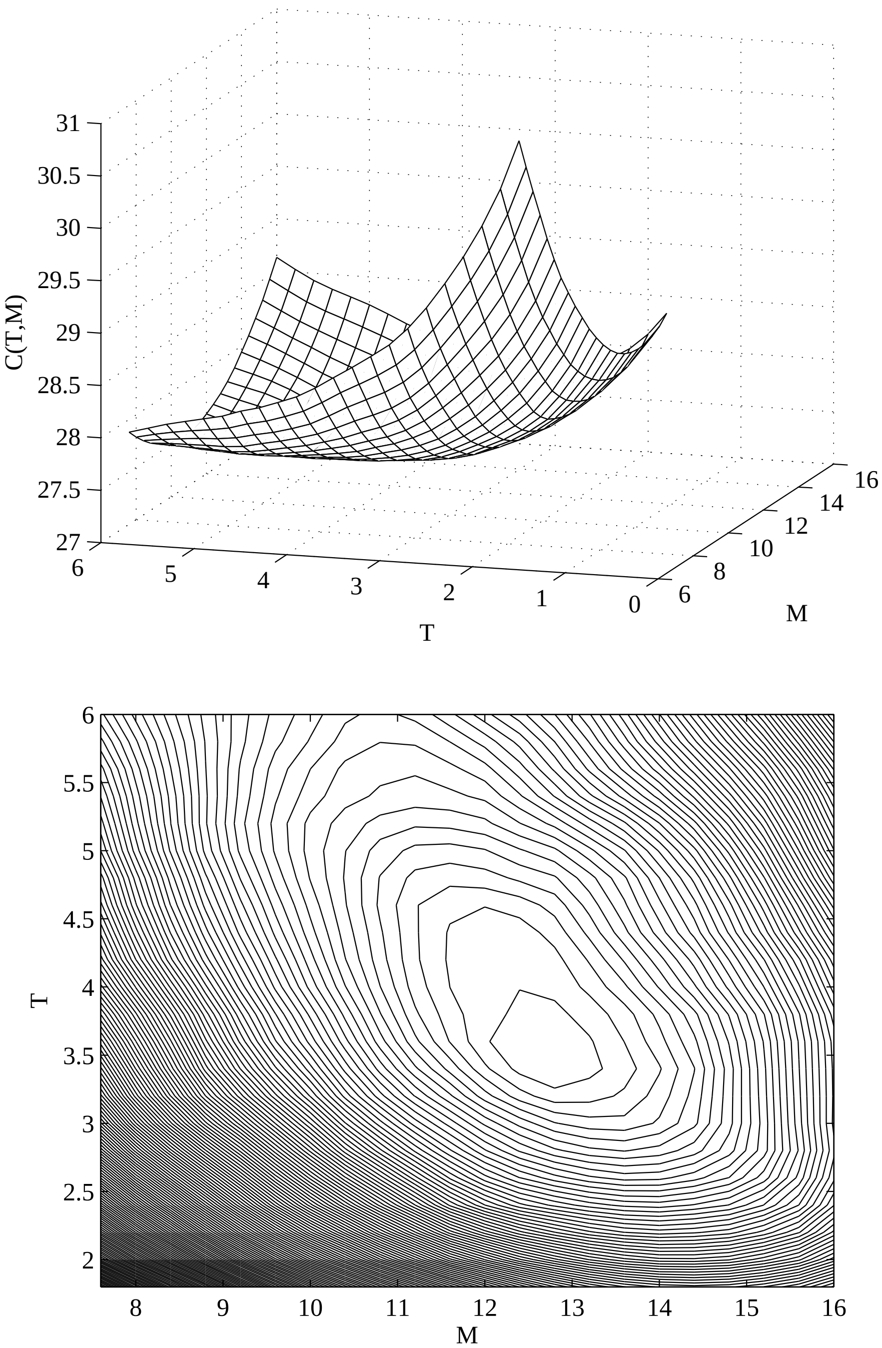

Fig. 2: Mesh and contour plots of $C(T, M)$.

	$\beta_{(-10\%)}$	$\beta_{(-5\%)}$	$\beta_{(-1\%)}$	β	$\beta_{(1\%)}$	$\beta_{(5\%)}$	$\beta_{(10\%)}$
$\alpha_{(-10\%)}$	1.3154	1.7764	4.6644	5.4867	5.6558	8.6687	10.6825
$\alpha_{(-5\%)}$	3.6085	0.4349	1.9725	2.8729	3.1102	5.6672	9.0845
$\alpha_{(-1\%)}$	5.6754	2.3691	0.1224	0.4699	1.5877	3.5827	6.2433
α	5.9838	2.6031	0.3590	0.0000	0.6763	2.9193	5.9754
$\alpha_{(1\%)}$	6.3446	3.4205	1.3157	0.2215	0.2270	2.5415	5.5062
$\alpha_{(5\%)}$	8.4384	5.2887	2.4793	2.2482	1.4543	0.7338	4.0128
$\alpha_{(10\%)}$	11.1060	7.6585	4.8535	4.4031	3.8389	1.4492	1.2767

Table 1: Relative variation percentages for the gamma process parameters.

	$b_{(-10\%)}$	$b_{(-5\%)}$	$b_{(-1\%)}$	b	$b_{(1\%)}$	$b_{(5\%)}$	$b_{(10\%)}$
$a_{(-10\%)}$	6.1626	7.4020	9.0060	9.1608	9.4140	10.5111	11.9452
$a_{(-5\%)}$	1.7469	2.6333	4.2221	4.3998	4.5034	5.4340	6.6690
$a_{(-1\%)}$	1.0238	0.2377	0.4463	0.7629	1.0334	1.7270	3.0284
a	2.2533	1.2188	0.3741	0.0000	0.2237	0.5311	1.6463
$a_{(1\%)}$	2.9492	2.2567	1.5155	0.8046	0.9630	0.2215	0.9767
$a_{(5\%)}$	5.9057	5.1229	3.9074	4.0102	4.0301	3.7158	2.7789
$a_{(10\%)}$	8.7546	8.2092	7.9796	8.2751	7.2229	7.1153	6.3340

Table 2: Relative variation percentages for the power-law intensity parameters.